

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: B | Time: 3 Hours | Max Marks: 128 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:

1. Answer all questions.
2. Figures to the right indicate full marks.

SECTION A: MCQs

1. Which of the following is listed as a type of scenario used in architectural evaluation? **[3 Marks]**
 - a) Budget scenario
 - b) Compliance scenario
 - c) Growth scenario
 - d) Performance scenario
2. According to the text, which of these is a formal and structured evaluation approach? **[3 Marks]**
 - a) SWOT Analysis Method
 - b) ATAM (Architecture Trade-off Analysis Method)
 - c) Agile Evaluation Method
 - d) Waterfall Evaluation Protocol
3. When using an off-the-shelf component like PayPal for online payments, what is a direct consequence for the system's architecture? **[3 Marks]**
 - a) Faster development of the PayPal component.
 - b) The system design must fit PayPal's API flow.
 - c) Unlimited customization of PayPal's internal behaviour.
 - d) Reduced need for architectural documentation.

SECTION B: SHORT ANSWERS

4. Differentiate between "Risks" and "Sensitivity points" as identified during a successful architectural evaluation. **[6 Marks]**

5. List any four key stakeholders whose involvement is crucial for successful architectural evaluation, as specified in the text. **[6 Marks]**
6. Explain "Faster development" as an impact of using off-the-shelf components, providing the example given in the source text. **[6 Marks]**
7. Name any four factors that architects consider when making design decisions for a software system. **[6 Marks]**
8. How do "Budget and Schedule" influence design decisions in software architecture? Provide an example from the text. **[6 Marks]**
9. Describe the impact of "Limited customization" when incorporating an off-the-shelf component, including the example provided in the text. **[6 Marks]**
10. Name the two specific formal and structured evaluation approaches mentioned in the text that improve evaluation success. **[6 Marks]**

SECTION C: LONG ANSWERS

11. Discuss how the expertise of an "Experienced Evaluation Team" contributes to a successful architectural evaluation. What specific understandings should such professionals possess according to the text? **[11 Marks]**
12. Explain the significance of "Alignment with Business Goals" in the successful evaluation of architecture. What three specific aspects must the architecture support to achieve this alignment? **[11 Marks]**
13. Detail the three different types of scenarios that are essential for good architectural evaluation. Briefly explain what each type describes. **[11 Marks]**
14. Describe the importance of "Non-Functional Requirements (Quality Attributes)" in design decision-making for software architecture, providing the specific example mentioned in the text. **[11 Marks]**
15. Discuss "Vendor dependency" and "Hard to replace later" as two significant impacts of using off-the-shelf components. Provide the examples given in the text for each impact. **[11 Marks]**
16. Enumerate and briefly explain any five factors that contribute to the successful evaluation of a software architecture. **[11 Marks]**
17. Explain the "Architecture must adjust to the component" and "Extra integration work" impacts when incorporating off-the-shelf components into a system. Include the examples provided for each impact. **[11 Marks]**
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